

Comparative Analysis of Deep Learning Models for Long-Term Electricity Demand Forecasting in Bangladesh Using Web-Scraped Data

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Abstract—As energy demand continues to rise in Bangladesh, there is a growing need for more accurate forecasting methods to improve the balance between electricity supply and consumption. Despite increased generation capacity, the country still experiences frequent disruptions due to limitations in prediction accuracy and structural inefficiencies within the power system. This research carries out an evaluative comparison of notable deep learning (DL) frameworks, including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional LSTM (BiLSTM), and Bidirectional GRU (BiGRU), for forecasting daily peak electricity demand at both the national level and across the country's eight divisions. The dataset was compiled from the Bangladesh Power Development Board (BPDB) using an automated web scraping pipeline. All models were trained on four years of historical data and evaluated using a one-year testing set. Among the models assessed, the BiGRU architecture outperformed others, achieving the lowest testing Mean Absolute Percentage Error (MAPE) value of 4.75%. The BiGRU model was also employed for division-wise forecasting, effectively capturing regional demand variations. Additionally, it was applied to unseen future dates, which are not included in the dataset, where it recursively predicted energy demand one day at a time and achieved an MAPE of 7.3%, demonstrating strong generalization capability. These results signify the aptitude of deep learning-based methodologies for enabling resilient and scalable energy consumption modeling.

Keywords— *Electricity demand forecasting, long-term load prediction, deep learning, LSTM, BiLSTM, GRU, BiGRU, web-scraped data, regional energy modeling, Bangladesh power system*

I. INTRODUCTION

Accurate forecasting of electricity demand is essential for maintaining grid stability, reducing load shedding, and improving operational efficiency, particularly in developing nations like Bangladesh, where frequent power outages disrupt both economic and social activities. Despite a 125% increase in installed generation capacity between 2016 and 2024, the country still experiences major mismatches between supply and demand due to poor forecasting, high transmission losses, and infrastructural limitations. Load shedding—the deliberate shutdown of electricity in specific areas to prevent grid collapse—remains common even though the country possesses sufficient generation capacity. For instance, in October 2024, peak demand reached 17,200 megawatts (MW), while the reserve capacity stood at 27,740 MW, indicating that Bangladesh had more than enough theoretical capacity to meet demand. However, due to the absence of accurate load forecasting, actual generation often either falls

short or exceeds real-time demand, resulting in unnecessary outages or wasted energy, both of which contribute to financial inefficiencies in the power sector [1].

Traditional statistical and rule-based forecasting models often fail to provide an idea of the complex, nonlinear, and temporally dependent patterns in electricity consumption, which are influenced by a variety of external factors such as weather, holidays, and economic activity [2]. On the other hand, Machine Learning (ML) and Deep Learning (DL) techniques have shown significant promise for time-series forecasting tasks due to their ability to learn long-term dependencies and handle sequence data effectively [3], [4].

In this paper, we devised distinct DL-based models for forecasting daily peak electricity demand across Bangladesh and its eight administrative divisions. Using over five years of historical electricity generation and demand data extracted from Bangladesh Power Development Board (BPDB) daily reports [5], we developed an automated pipeline for data collection. The data then underwent preprocessing and feature engineering. The incorporation of temporal, environmental, and holiday-related features further enhanced forecasting precision. We then trained and evaluated several DL architectures to assess forecasting accuracy on both the national and divisional levels.

II. LITERATURE REVIEW

In recent years, ML and DL methods have increasingly supplanted traditional forecasting models in energy demand prediction due to their superior ability to handle complex, nonlinear, and highly variable load patterns. Among these, Recurrent Neural Networks (RNNs)—notably Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and their hybrid variants—have gained prominence for their ability to capture sequential dependencies effectively. This shift reflects a growing recognition that conventional models such as Autoregressive Integrated Moving Average (ARIMA), Support Vector Machine (SVM), and regression-based approaches often fail to represent the temporal and contextual complexities inherent in electricity consumption, especially under the influence of external factors like weather fluctuations, holidays, and economic activities.

Significant progress has been made globally through hybrid deep learning architectures. Rafi et al. introduced an integrated Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM) model for short-term load forecasting, demonstrating its superiority over standalone

LSTM, Radial Basis Function Networks (RBFN), and Extreme Gradient Boosting (XGBoost) models. The model achieved Mean Absolute Percentage Error (MAPE) values consistently below 5%, highlighting the benefit of CNN layers in extracting localized temporal features that enhance LSTM's sequential learning capabilities [6]. Amalou et al. further validated the robustness of GRU architectures, reporting improved computational efficiency and stability over LSTM and vanilla RNN models across multiple forecast horizons [7]. In another study, Alhussein et al. developed a CNN-LSTM hybrid model for short-term household electricity load forecasting using Australian residential data. Their model outperformed a baseline LSTM by 4.01% to 5.98% in accuracy across 1 to 6 time steps, although the average MAPE remained high at 40.38% due to the complexity of household-level demand [8].

Within the context of Bangladesh, deep learning models have also been explored extensively, though with limitations in geographic granularity and feature diversity. Miah et al. developed a Bidirectional LSTM (BiLSTM) model trained on hourly national load data from Power Grid Company of Bangladesh (PGCB), obtaining a Mean Absolute Error (MAE) of 376.72 MW and a Mean Squared Error (MSE) of 340,494.74, illustrating the model's efficacy in capturing bidirectional dependencies in time series data [9]. Islam et al. focused on Chattogram city, implementing an LSTM model that yielded an MAE of 92.45 MW and Root Mean Squared Error (RMSE) of 127.68 MW, reducing forecasting errors by over 50% compared to SVM baselines [10]. Baul et al. proposed a CNN-GRU hybrid model capable of forecasting across nine power zones. Their model outperformed CNN-LSTM, GRU, and Transformer models in most regions, though the predictive contribution of weather features was found to be limited [11]. Islam et al. enhanced forecast precision by developing a hybrid LSTM-Prophet model that incorporated exogenous features including Gross Domestic Product (GDP), temperature, humidity, and population. The model achieved a MAPE of 2.3% for January–February 2024 only, underscoring the impact of comprehensive feature engineering [12].

Other Bangladesh-centric studies have advanced architectural innovations. Akhter et al. introduced a CNN-BiLSTM model for Dhaka's load data, achieving MAPE of 1.64%, RMSE of 0.122, and MAE of 0.092—among the most accurate results reported for urban electricity demand [13]. Al-Amin et al. proposed a parallel CNN-BiLSTM structure that successfully captured both short-term fluctuations and seasonal effects, achieving MAPE of 1.414 and RMSE of 229.78 over six years of data including weather and holiday indicators [14]. Biswas et al., in contrast, employed a tree-based ensemble model using Light Gradient Boosting Machine (LightGBM) for one-day-ahead load forecasting in Dhaka, achieving a MAPE of 5.08%. While the model demonstrated good short-term accuracy, it lacked the ability to learn sequential temporal dependencies inherent in time-series data, limiting its suitability for multi-day forecasting scenarios [15].

Despite these advancements, several critical limitations persist. First, most studies validate their models using temporally adjacent datasets—such as training on one year and testing on the next—which inflates accuracy metrics and undermines model robustness on truly unseen future data. Longitudinal evaluation, where models are trained on data

from multiple past years (e.g., 2020–2023), validated on an intermediate year (e.g., 2024), and tested on completely unseen future data (e.g., 2025), remains virtually absent in current literature. Second, the geographical focus remains narrowly restricted to major urban areas like Dhaka and Chattogram, with few models extending coverage to all eight administrative divisions. Third, while exogenous variables such as temperature and GDP are occasionally used, more nuanced contextual features—such as regional holiday calendars, religious events, and industrial activity—are either ignored or insufficiently integrated. Fourth, Transformer-based architectures, despite their theoretical advantages, have shown inconsistent performance in the Bangladeshi context, as evidenced by Baul et al., due to higher overfitting risk and computational demands [11].

Finally, reproducibility is hindered by the limited availability of well-structured, open-access datasets. Most studies rely on closed or non-standardized data sources, making comparative benchmarking difficult. Very few provide region-wise performance metrics, further limiting insights into model generalizability and equity across diverse geographic settings.

In summary, while the literature demonstrates significant progress in the application of deep learning to electricity demand forecasting, existing works fall short in terms of generalizability, regional coverage, and real-world deployment readiness. These gaps highlight the urgent need for a robust, granular, and reproducible forecasting framework that incorporates localized external variables and is evaluated using temporally disjoint data—capable of informing policy and operations across the full spectrum of Bangladesh's regional energy ecosystem.

III. DATA COLLECTION AND PREPROCESSING

A. Data Extraction and Compilation

To construct the dataset, we implemented a fully automated web scraping pipeline to systematically collect daily electricity generation reports from the BPDB website, covering the period from 21 November 2019 to 31 December 2024 [5]. Using Python-based tools, the script extracted the relevant PDF URLs and systematically retrieved the files. From each report, only the relevant section containing national and divisional electricity demand and generation metrics was extracted using the *pdfplumber* library. The extracted data included attributes such as the actual date of the report, day of the week, maximum demand at evening peak (both generation end and sub-station end), highest and minimum daily generation, day-peak and evening-peak generation, and minimum generation forecast. In addition, environmental and operational constraints were captured, including maximum temperature in Dhaka, gas and liquid fuel (LF) limitations, coal supply limitations, low water level in Kaptai Lake, and details of plants under shutdown or maintenance. The dataset also incorporated division-wise electricity demand, supply, and load shedding for Dhaka, Chattogram, Rajshahi, Khulna, Mymensingh, Rangpur, Barishal, and Sylhet. The processed data were structured and saved in Excel format for subsequent analysis and have been submitted to Data in Brief journal, which is under review for potential publication [16].

B. Missing Value Handling

Once the dataset was compiled, several preprocessing steps were performed. As the dataset represents a time series,

missing values were handled using interpolation based on the preceding and succeeding data points to maintain continuity.

C. Feature Engineering

To improve model accuracy in capturing variations in electricity consumption, a holiday feature was engineered by categorizing days into single-day holidays (e.g., national holidays), continuous holidays (e.g., Eid vacations, long weekends), and special events (e.g., religious festivals), enabling the model to distinguish between regular and exceptional demand patterns. Additionally, the data was lagged by one day to capture short-term temporal trends and recent consumption behavior.

D. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to uncover underlying patterns, summarize the dataset, explore correlations between features, and gain a deeper understanding of the dataset's structure. To examine long-term and seasonal variations in electricity demand, Maximum demands at evening peak were plotted over a five-year period. As shown in Fig. 1, the analysis reveals a steadily increasing trend in peak electricity demand, along with seasonal patterns over time. Electricity consumption varied significantly between holidays and regular days. As illustrated in Fig. 2, certain holidays exhibited a sharp decline in demand, while others, particularly major festive periods, showed increased consumption. The holiday categories are defined as follows: 0 represents no holiday, 1 indicates a single-day holiday, 2 corresponds to continuous holidays such as Eid or Puja, and 3 denotes special religious occasions like Ramadan. The analysis reveals that energy demand tends to increase during Ramadan compared to regular days.

Further analysis examined average daily electricity consumption across a week as shown in Fig. 3. The analysis of weekend vs. weekday trends in power demand reveals a noticeable dip on Fridays, with the lowest mean demand at 10,920 MW, indicating reduced industrial and commercial activity. In contrast, Tuesdays recorded the highest mean demand at 11,819 MW, reflecting increased weekday electricity consumption. Monthly demand trends for the entire country were visualized in Fig. 4. The monthly electricity demand analysis reveals clear seasonal variations, with demand peaking in April, May, and September, likely due to increased cooling requirements during the summer months. Conversely, demand drops in January, February, and November, reflecting reduced energy consumption during the winter. Additionally, Fig. 5 presents region-wise monthly demand, uncovering substantial spatial variation in electricity consumption across different divisions. Regional monthly trends indicate that Dhaka, Chattogram, and Khulna consistently exhibited the highest power consumption levels throughout the year.

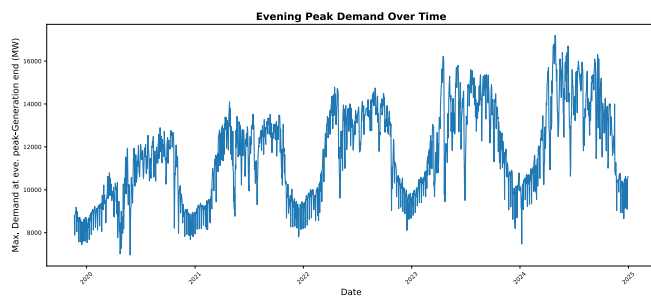


Fig. 1. Temporal variation of evening peak electricity demand in Bangladesh from 2019 to 2024.

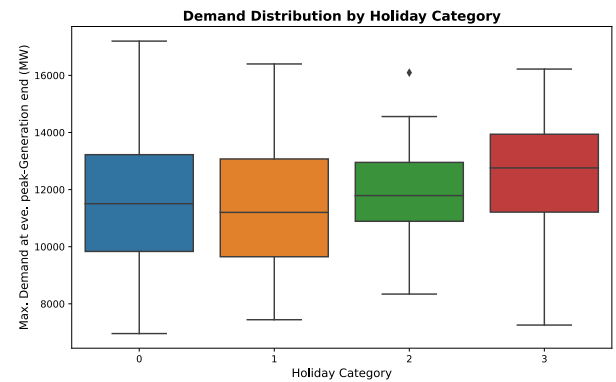


Fig. 2. Distribution of evening peak electricity demand across different holiday categories.

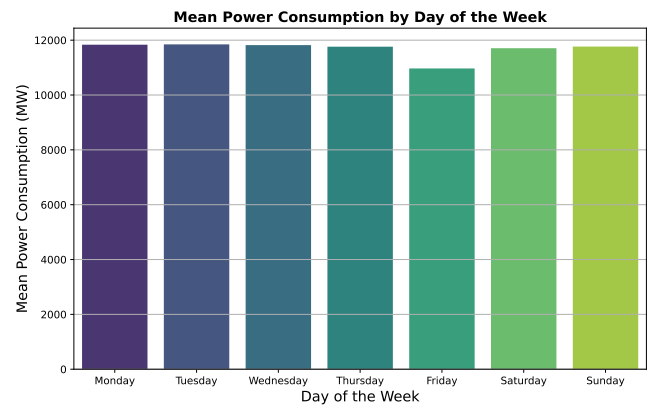


Fig. 3. Average power consumption across days of the week.

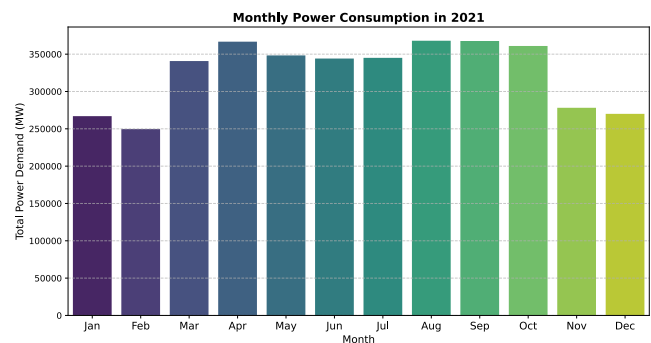


Fig. 4. Monthly total electricity consumption in 2021.

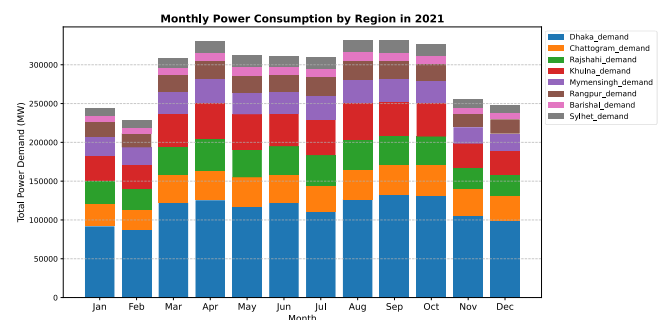


Fig. 5. Monthly electricity consumption by region in 2021.

E. Feature Selection

Our target was to predict maximum demand at a particular time for the whole country and later for each region.

1) *Target features*: To predict complete demand for the whole country we used Max. Demand at Evening Peak (Generation End) data from the BPDB as our target feature. For predicting regional demand, we chose demand data available for each division.

2) *Input Feature*: Based on the relationship between our target features and other available features in our dataset, we chose several features by statistical analysis as well as by feature elimination. The features selected from available features are maximum temperature in Dhaka, year, month, day, weekday, holiday. Additional engineered lag features such as previous day's electricity demand and temperature was also included in the input feature set.

IV. MODEL DESIGN

RNN-based architectures have proven effective in time-series forecasting owing to their strength in learning temporal dependencies in sequential datasets. For electricity demand forecasting, the four models, namely LSTM, BiLSTM, GRU, BiGRU that we are considering are each selected for their demonstrated efficacy in handling sequence data with varying degrees of complexity and computational efficiency.

A. Architectures for Time-Series Forecasting

LSTM networks introduce memory cells and gated mechanisms that allow selective retention and forgetting of information across time steps. This architecture eliminates the vanishing gradient problem that occurs in traditional RNNs, making it suitable for modeling long-term patterns in time-series data. As shown by Hochreiter et al. [17] and later confirmed in applied forecasting domains, LSTM models consistently outperform traditional approaches when learning from long temporal contexts [3]. While effective, the increased number of parameters results in longer training times and higher memory usage. While the model demonstrated good short-term accuracy, it lacked the ability to learn sequential temporal dependencies inherent in time-series data, limiting its suitability for multi-day forecasting scenarios.

The GRU architecture simplifies LSTMs by replacing the separate forget and input gates with a single update gate, resulting in more efficient training while preserving performance in sequence learning tasks. Studies comparing GRU and LSTM architectures often report similar levels of accuracy, with GRUs offering the advantage of reduced computational demand [4].

Bidirectional variants of both LSTM and GRU extend their predictive capability by processing each input sequence in both forward and backward directions. This configuration allows the model to incorporate information from both past and future time steps within a given context window, which is especially useful in domains with strong periodic patterns or delayed effects. Bidirectional recurrent networks have demonstrated improved performance in forecasting applications where temporal symmetry and future context contribute significantly to the prediction [18].

B. Model Configuration

As input, the models take a multivariate time series composed of both numerical and categorical variables, including the previous day's peak electricity demand, the previous day's temperature, the current day's maximum temperature, weekday index, holiday type, day, month, and year. These features are commonly used in load forecasting literature due to their high explanatory power and are prepared using standardization with the StandardScaler, which centers the data by removing the mean and scales it to unit variance, allowing stable training and faster convergence. Input data are

structured into fixed-length sliding windows, enabling the recurrent networks to learn patterns from recent temporal histories.

All hidden layers are activated using the Rectified Linear Unit (ReLU) function to introduce non-linearity. ReLU offers computational simplicity and avoids the saturation effects of sigmoid or tanh, making it a reliable choice for deep and recurrent architectures. The output layer also uses a ReLU activation function to support unbounded real-valued predictions.

Regularization is implemented using two complementary strategies. Dropout is applied after recurrent layers, with rates between 0.1 and 0.3, to randomly deactivate neurons during training and reduce overfitting. L2 weight regularization is introduced in inner layers to penalize large weights and encourage smoother solutions, both methods being widely adopted in deep learning to improve generalization. Another preventive measure taken for overfitting is the application of the EarlyStopping function. By utilizing it, the training of the models was stopped once the validation loss had no significant change despite going through a set number of epochs.

The unidirectional LSTM and GRU models are built with one or two recurrent layers containing 32 to 128 hidden units, followed by dense layers. Their bidirectional counterparts, BiLSTM and BiGRU, utilize two bidirectional recurrent layers, each with 32 to 128 units per direction, and concatenate the forward and backward outputs before passing them through fully connected layers. All model variants conclude with a single-neuron output layer for peak demand prediction.

Adam optimization is employed during training for its ability to adjust learning rates individually and perform reliably in non-stationary, sparse-gradient contexts. The loss function is MSE as in (1), chosen for its sensitivity to large deviations, while model performance is evaluated using MAE as well as MAPE, as given in (2) and (3), due to their interpretability in the unit scale of demand values.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (3)$$

Where-

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of observations

V. EXPERIMENTAL SETUP

To compare the performance of DL models in long-range electricity demand prediction, we employed a well-structured experimental setup encompassing sequence modeling and hyperparameter optimization. Each model was evaluated independently to ensure an unbiased assessment of its forecasting capability. The models are constructed using Tensorflow.keras.Sequential with GRU/LSTM layers followed by dense layers.

TABLE I. TUNED HYPERPARAMETER FOR INDIVIDUAL MODELS

Parameters	Model Name			
	<i>LSTM</i>	<i>GRU</i>	<i>BiLSTM</i>	<i>BiGRU</i>
Recurrent layers	2	2	2	2
Unit 0	96	96	128	96
Unit 1	64	64	96	64
Dense unit	56	56	48	56
Learning rate	6.684×10^{-4}	6.684×10^{-4}	4.489×10^{-4}	6.684×10^{-4}

The supervised learning format is created using a fixed sequence window of 10 days. Each input sample consists of 10 consecutive days of historical features, and the model is trained to predict the peak demand for the 11th day. This results in an input shape of (samples, 10, features) and supports temporal modeling across all recurrent layers.

The data is split chronologically into training and testing subsets using an 80:20 split, ensuring no information from the future influences the training process. To ensure reproducibility of results, a random seed of 42 is set using Python's random, NumPy, and TensorFlow libraries.

Model selection and optimization are conducted using Bayesian Optimization with the Keras Tuner library. The tuner explores hyperparameters including the number of recurrent layers (1 to 3), number of units per layer (32 to 128), dense layer units (8 to 64), and learning rate (log-uniform from $1e-5$ to $1e-3$). The tuning process minimizes validation MAE over 10 trials with 2 executions per trial to reduce noise. The optimized hyperparameter values from each model are provided in Table I.

To reduce overfitting, L2 regularization is applied to the dense layers, with tuning values ranging from 0.0005 to 0.01. Additionally, dropout is used with rates between 0.1 and 0.5, depending on the model complexity.

Early training termination is controlled by early stopping, configured with a patience of 10 epochs, and the best model weights are restored once validation loss stops improving. To enhance convergence, a learning rate scheduler (ReduceLROnPlateau) is used, decreasing the learning rate by a factor of 0.5 after 10 epochs without validation loss improvement. These mechanisms improve convergence near optima and reduce unnecessary computation.

All models are trained using the Adam optimizer, with the loss function set to MSE. MAE and MAPE are recorded for evaluation. Training is conducted for up to 200 epochs using a batch size of 64, though most runs converge earlier due to early stopping.

VI. RESULT AND ANALYSIS

The forecasting performance of various models for nationwide energy demand was evaluated using MAE and MAPE. Among the models, the BiGRU model outperformed LSTM, BiLSTM, and GRU, demonstrating the highest accuracy. Consequently, the BiGRU model was employed for division-wise forecasting. The BiGRU model was also used for long-term energy demand forecasting across the entire country.

A. Comparative Evaluation of Different Models for Nationwide Load Forecasting

A comparison of the four models demonstrated in Table II revealed that BiGRU provided the most accurate and reliable forecasts, with the lowest testing MAPE of 4.75%. While GRU and BiLSTM showed moderate improvements, LSTM underperformed with a high testing MAPE of 5.55%. The actual versus predicted electricity demand plots for LSTM, BiLSTM, GRU, and BiGRU models are shown in Fig. 6, Fig. 7, Fig. 8, and Fig. 9 respectively. These visualizations highlight the improved alignment of BiGRU predictions with actual demand, particularly in capturing peak fluctuations and long-term seasonal trends.

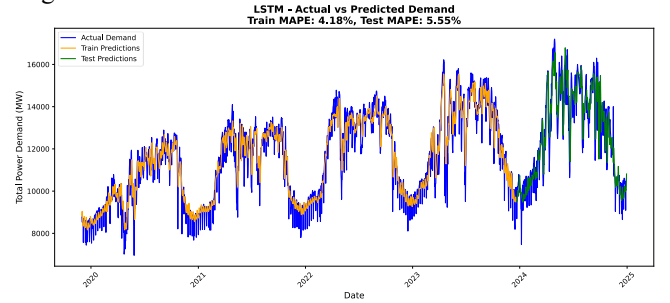


Fig. 6. Actual vs. Predicted Daily Electricity Demand Using the LSTM Model.

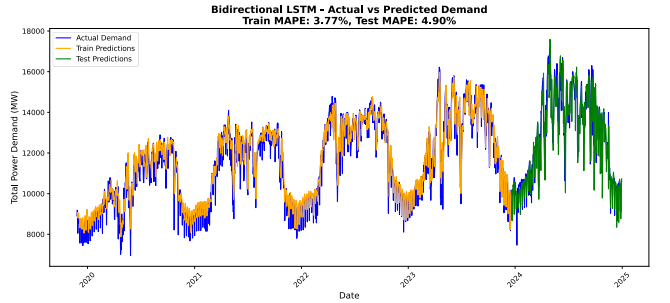


Fig. 7. Actual vs. Predicted Daily Electricity Demand Using the BiLSTM Model.

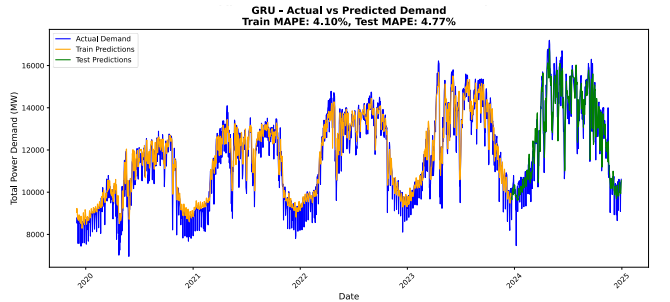


Fig. 8. Actual vs. Predicted Daily Electricity Demand Using the GRU Model.

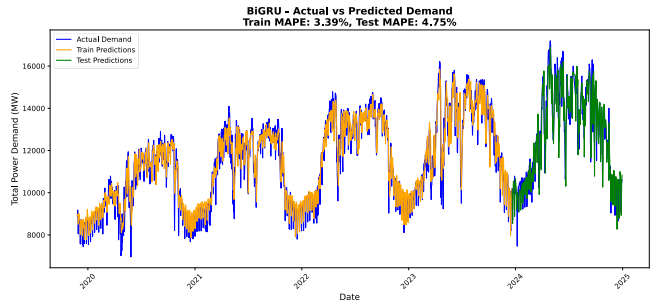


Fig. 9. Actual vs. Predicted Daily Electricity Demand Using the BiGRU Model.

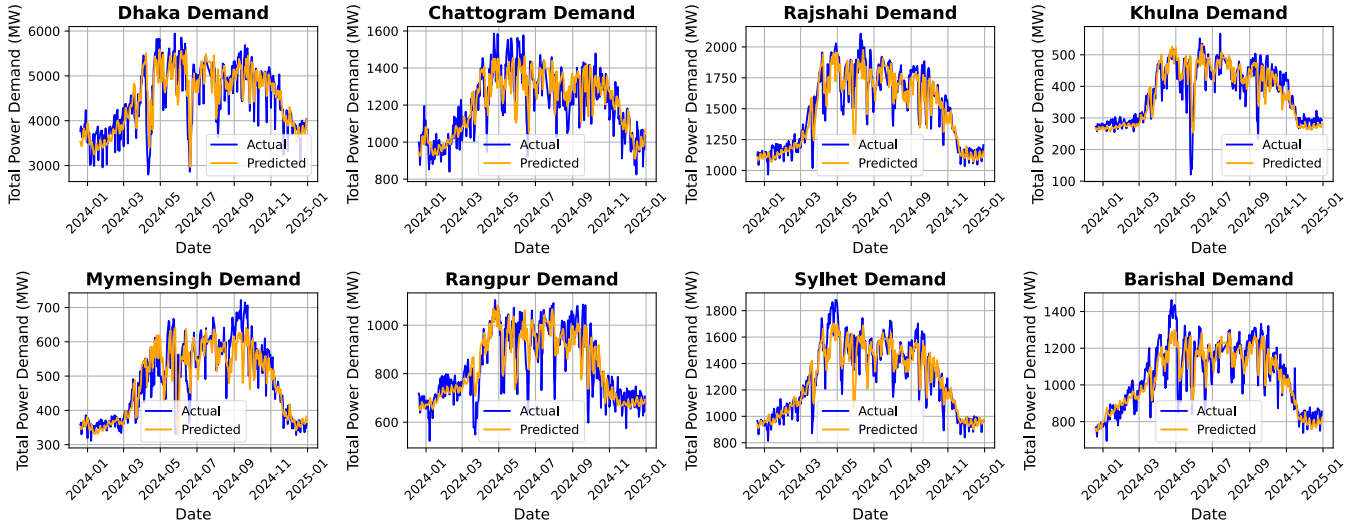


Fig. 10. Division-wise actual vs. predicted electricity demand using the BiGRU model for 2024.

TABLE II. COMPARISON OF LSTM, BiLSTM, GRU, AND BiGRU MODELS BASED ON MAE, MAPE, AND ACCURACY

Model Name	Performance Metrics					
	MAE (MW)		MAPE (%)		Accuracy (%)	
	Train	Test	Train	Test	Train	Test
LSTM	459.96	696.87	4.18	5.55	95.82	94.45
BiLSTM	410.45	624.10	3.77	4.90	96.23	95.1
GRU	444.43	598.46	4.10	4.77	95.9	95.23
BiGRU	374.32	595.95	3.39	4.75	96.61	95.25

TABLE III. DIVISION-WISE PERFORMANCE METRICS OF THE BiGRU MODEL FOR ELECTRICITY DEMAND FORECASTING

Division Name	Performance Metrics					
	MAE (MW)		MAPE (%)		Accuracy (%)	
	Train	Test	Train	Test	Train	Test
Dhaka	195.5	281.14	5.37	6.66	94.63	93.34
Chattogram	58.76	73.92	5.35	6.2	94.65	93.8
Rajshahi	54.88	73.87	4.75	5.63	95.25	94.37
Khulna	63.73	84.96	4.84	5.57	95.16	94.43
Mymensingh	44.36	63.03	5	6	95	94
Rangpur	40.24	55.24	6.5	6.87	93.5	93.13
Barishal	17.53	29.4	5.55	9.01	94.45	90.99
Sylhet	25.41	37.61	5.83	7.57	94.17	92.43

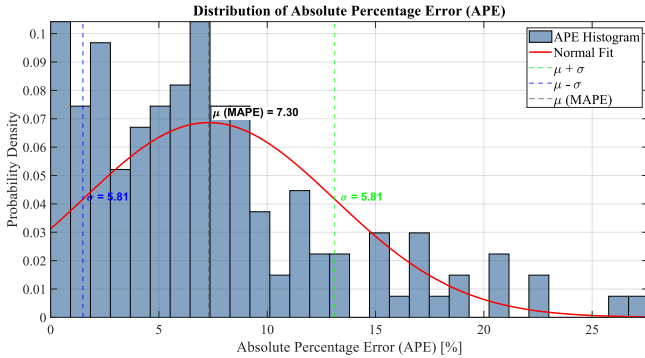


Fig. 11. Distribution of APE for BiGRU model's long term forecast from 31 December 2024 to 27 May 2025.

B. Division-Wise Energy Demand Forecasting Using the BiGRU Model

Fig. 10 illustrates the electricity demand forecasts over a one-year testing period starting from 1 January 2024 until 30 December 2024, for Bangladesh's eight divisions: Dhaka, Chattogram, Rajshahi, Khulna, Mymensingh, Rangpur, Sylhet and Barishal. This one-year period was part of the dataset and was reserved exclusively for testing to evaluate the generalization performance of the BiGRU model, which had previously been identified as the most accurate during nationwide evaluation. Model performance varied across divisions: in some cases, the predicted values closely aligned with actual demand, while in others, noticeable deviations were observed. The MAE and MAPE for both training and testing across all divisions are summarized in Table III.

C. Long-Term Energy Demand Forecasting for Bangladesh Using the BiGRU Model

Using the BiGRU model, we implemented a recursive prediction algorithm to enable the long-term forecast of electricity demand for future unseen dates that are not present in the dataset. Starting from the last available date in the data, the model utilizes a sliding window of the previous five days' features, including peak demand, temperature, and holiday indicators, to generate the forecast for the next day. Each predicted value is subsequently incorporated into the input sequence, allowing the model to iteratively forecast one day at a time until the target date is reached. As actual future observations are unavailable, historical temperature data from the corresponding day of the previous year is used to approximate future conditions. The output is a DataFrame containing predicted daily peak demands up to the specified target date. To validate the predictions, we extracted actual demand data from the BPDB website and compared it against the model outputs. Fig. 11 shows the distribution of Absolute Percentage Error (APE) for the long-term forecast using the BiGRU model over the period from 31 December 2024 to 27 May 2025. The histogram indicates that most predictions fall within a narrow APE range, while the red line represents the normal fit curve. The calculated MAPE is approximately 7.3%, with a standard deviation of 5.81%, reflecting the model's stable performance across the 148-day forecast horizon. The percentage error between the predicted and actual values of different dates within this period ranges from 0.0143% to 27.42%.

TABLE IV. COMPARISON OF ACTUAL AND PREDICTED ELECTRICITY DEMAND ON RANDOMLY SELECTED DATES

Date	Actual Electricity Demand (MW)	Predicted Electricity Demand (MW)	APE (%)
31 December 2024	10381	10606	2.16
1 January 2025	10337	10388	0.49
15 January 2025	10893	10340	5.08
1 February 2025	11386	10068	11.57
18 February 2025	12212	11371	6.88
3 March 2025	12566	12295	2.15
17 March 2025	14493	14700	1.43
1 April 2025	12576	11980	4.74
19 April 2025	14500	14031	3.24
2 May 2025	13677	13214	3.38
15 May 2025	14921	14878	0.28
27 May 2025	16420	15809	3.72

For additional validation, Table IV provides a detailed comparison of predicted and actual values on randomly selected dates within this period.

VII. CONCLUSION

This study introduces a scalable deep learning framework for forecasting daily peak electricity demand and regional load shedding patterns across Bangladesh. By leveraging a multivariate feature set—including holiday categorization, weather conditions, and lagged demand signals—our models demonstrated robust performance across eight administrative divisions. The BiGRU architecture, in particular, achieved the lowest forecast error, effectively capturing temporal dependencies and regional variability. Our forecasting approach extended beyond traditional evaluation boundaries, generating day-by-day predictions up to May 2025 on entirely unseen data, thus emphasizing the model's real-world applicability and generalizability. Bangladesh incurs over USD 1 billion annually in capacity payments and subsidies due to inefficient planning and surplus reserve margins [1]. Improving load forecasting and reducing load shedding could save approximately USD 1.2 billion annually. By minimizing forecasting errors, our model contributes directly to reducing these losses and enabling smarter, demand-responsive energy resource allocation.

Future work can focus on developing a high-resolution, publicly accessible hourly electricity demand dataset, enabling finer-grained predictions and more dynamic load management. Additionally, incorporating advanced architectures such as Transformer-based temporal models and hybrid attention mechanisms may further improve forecasting fidelity. These advances will help build adaptive energy systems that support sustainable and equitable power distribution in developing economies.

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